

Algebra

Class -1

Q. If $x^2 + 1/x^2 = 29$, then find the value of $x - 1/x$?

(a) ± 3 (b) $\pm 4\sqrt{3}$ (c) ± 4 (d) $\pm 3\sqrt{3}$

Q. If $x + 1/x = 1$, then what is the value of $(x^2 + 7x + 1)/(x + 11x + 1)$?

(a) $1/3$ (b) $2/3$ (c) $3/4$ (d) $1/4$

Q. If $x + 1/x = 7$ then find the value of $x^2 + 1/x^2$?

(a) 49 (b) 51 (c) 5 (d) 47

Q. If $2x + \frac{2}{x} = 5$, then, the value of $x^3 + \frac{1}{x^3} + 2$ will be ?

(a) $\frac{71}{8}$ (b) $\frac{91}{11}$ (c) $\frac{81}{7}$ (d) $\frac{81}{8}$

Q. If $x + \frac{1}{x} = \sqrt{3}$ then the value of $(x^6 + \frac{1}{x^6})$ is ?

(a) -2 (b) 2 (c) -6 (d) 6

If $\left(x^2 + \frac{1}{x^2}\right) = 7$, and $0 < x < 1$, find the value of $x^2 - \frac{1}{x^2}$.

यदि $\left(x^2 + \frac{1}{x^2}\right) = 7$, और $0 < x < 1$ है, तो $x^2 - \frac{1}{x^2}$ का मान ज्ञात कीजिये ।

a) $3\sqrt{5}$

b) $4\sqrt{3}$

c) $-4\sqrt{3}$

d) $-3\sqrt{5}$

24. If $x^2 - 7x + 1 = 0$, and $0 < x < 1$, what is the value of $x^2 - \frac{1}{x^2}$?

यदि $x^2 - 7x + 1 = 0$, और $0 < x < 1$, तो $x^2 - \frac{1}{x^2}$ का मान क्या है?

a) $21\sqrt{5}$

b) $-21\sqrt{5}$

c) $28\sqrt{5}$

d) $-28\sqrt{5}$

$X^4 + 1/X^4 = 14, X > 0,$
Find $2x$?

- A. $\sqrt{6} + \sqrt{2}$
- B. $\sqrt{6} - \sqrt{2}$
- C. 4
- D. 14

$X^{24} + 1/X^{24} = 7, x > 0,$
find $x^6 + 1/x^6$?

- A. $\sqrt{2}$
- B. 3
- C. $\sqrt{5}$
- D. $\sqrt{3}$

$$X^{124} + 1/X^{124} = 62, x > 0,$$

find $2x^{31}$?

$$\sqrt{10} + \sqrt{6}$$

$$\sqrt{10} + \sqrt{2}$$

$$\sqrt{10}$$

$$\sqrt{2}$$

$$X^4 + 16/X^4 = 433, x > 0,$$

find $x - 2/x$?

$$5$$

$$\sqrt{17}$$

$$3$$

$$\sqrt{15}$$

If $x > 0$, $x^4 + 1/x^4 = 2207$, find $x^5 + 1/x^5$?

15127

15134

15141

15130

If $p^4 = 4354 - 1/p^4$, find $p^3 - 1/p^3$?

436

536

416

516

**If $x^4 + 1/x^4 = 14159$,
Find $x + 1/x$?**

11

69

81

121

**If $X^4 + 1/x^4 = 727$, $x > 1$,
Find $x - 1/x$?**

2

3

4

5

**If $x^4 + 1/x^4 = 2599$,
 $x > 0$, find $x - 1/x$?**

4

5

6

7

**If $x^4 + 1/x^4 = 47$, $x > 0$,
find $(2x - 3)^2$?**

4

5

6

7

If $x^4 + 1/x^4 = 527$, $x > 0$, find $(2x - \sqrt{21})^2$?

4

5

6

7

**If $x^4 - 142x^2 + 1 = 0$, $x > 0$,
find x ?**

$3 - \sqrt{9}$

$6 - \sqrt{35}$

$6 + \sqrt{35}$

$4 + \sqrt{15}$

If $81x^4 + 1/x^4 = 3007$,
 $x > 0$,
find $(6x-7)^2$?

59

60

61

62

If $256x^4 + 1/x^4 = 752$,
 $x > 0$, find $4x + 1/x$?

4

5

6

7

If $x^4 + 1/x^4 = 1154$, ($x > 0$),
find $x + 1/x$?

$\sqrt{34}$

$\sqrt{32}$

6

18

If $x^{123} + 1/x^{123} = 702$, find
 x^{41} ?

A. $(9 + \sqrt{77})/2$

B. $(9 - \sqrt{77})/2$

C. $(9 + \sqrt{77})$

D. $(9 + \sqrt{77})$

If $x^4 - 1762x^2 + 1 = 0$, find $(2x - 4\sqrt{110} - 42)^2$?

0

1

2

3

If $7b - \frac{1}{4b} = 7$, then what is the value of $16b^2 + \frac{1}{49b^2}$?

यदि $7b - \frac{1}{4b} = 7$ है, तो $16b^2 + \frac{1}{49b^2}$ का मान क्या है?

a) $\frac{80}{49}$

b) $\frac{104}{7}$

c) $\frac{120}{7}$

d) $\frac{7}{2}$

If $x > 0$, and $x^4 + \frac{1}{x^4} = 254$, what is the value of $x^5 + \frac{1}{x^5}$?

यदि $x > 0$, और $x^4 + \frac{1}{x^4} = 254$, $x^5 + \frac{1}{x^5}$ का मान क्या है?

a) $717\sqrt{2}$

b) $723\sqrt{2}$

c) $720\sqrt{2}$

d) $726\sqrt{2}$

.If $a + \frac{1}{a} = 7$, then $a^5 + \frac{1}{a^5}$ is equal to:

यदि $a + \frac{1}{a} = 7$, फिर $a^5 + \frac{1}{a^5}$ के बराबर है:

a) 15127

b) 13127

c) 14527

d) 11512

If $\left(x - \frac{1}{x}\right) = \sqrt{6}$, and $x > 1$, what is the value of $\left(x^8 - \frac{1}{x^8}\right)$?

यदि $\left(x - \frac{1}{x}\right) = \sqrt{6}$, और $x > 1$, $\left(x^8 - \frac{1}{x^8}\right)$ का मान क्या है?

a) $1024\sqrt{15}$

b) $992\sqrt{15}$

c) $998\sqrt{15}$

d) $1012\sqrt{15}$

If $x + \frac{1}{x} = -6$, what will be the value of $x^5 + \frac{1}{x^5}$?

यदि $x + \frac{1}{x} = -6$ है, तो $x^5 + \frac{1}{x^5}$ का मान क्या होगा?

a) -7776

b) -6726

c) -6738

d) -6732

. If $x + \frac{1}{x} = 5$, and $x > 1$, what is the value of $\left(x^8 - \frac{1}{x^8}\right)$?

यदि $x + \frac{1}{x} = 5$, और $x > 1$, $\left(x^8 - \frac{1}{x^8}\right)$ का मान क्या है?

a) $60605\sqrt{21}$

b) $60615\sqrt{21}$

c) $60705\sqrt{21}$

d) $60725\sqrt{21}$

13. If $x + \frac{1}{x} = 2\sqrt{2}$, and $x > 1$, what is the value of $\left(x^6 - \frac{1}{x^6}\right)$?

यदि $x + \frac{1}{x} = 2\sqrt{2}$, और $x > 1$ है, तो $\left(x^6 - \frac{1}{x^6}\right)$ का मान क्या है?

a) $140\sqrt{2}$

b) $116\sqrt{2}$

c) $144\sqrt{2}$

d) $128\sqrt{2}$

16. If $x - \frac{1}{x} = 2\sqrt{2}$, and $x > 1$, what is the value of $x^6 - \frac{1}{x^6}$?

यदि $x - \frac{1}{x} = 2\sqrt{2}$, और $x > 1$, तो, $x^6 - \frac{1}{x^6}$ का मान क्या है?

a) $372\sqrt{6}$

b) $384\sqrt{6}$

c) $396\sqrt{6}$

d) $420\sqrt{6}$

17. Simplify the following expression.

निम्न अभिव्यक्ति को सरल बनाएँ।

$$[(1 + p)(1 + p^2)(1 + p^4)(1 + p^8)(1 + p^{16})(1 - p) - 1]$$

a) $-p^{32}$

b) p^{32}

c) $(1 + p^{32})$

d) $(1 - p^{32})$

Thankyou

Algebra

Class -2

If $x^2 + 1/x^2 = 6$, find $10x - 10/x$?

20 only

20 and -20 only

40 only

40 and -40 only.

If $x^2+4y^2=40$, $xy=6$,
 $x>2y$, find $x-2y$?

1

2

3

4

If $X=1+\sqrt{2}$ then find $\sqrt{X}+1/\sqrt{X}$?

2.1014

2.1973

1.9876

1.9996

If $a+2b=27$, $a^3+8b^3=5427$, find $2ab$?

156

149

172

176

**If $16x^2+y^2= 48$ and $xy = 2$,
 $x,y>0$, find $64x^3+y^3$?**

240

300

320

340

If $16x^4 + \frac{1}{16x^4} = 14159$, then find the value of $2x + \frac{1}{2x}$.

1. 9

2. 11

3. 12

4. 15

If $x + \frac{1}{x} = 2$, then find the value of $(x^4 + \frac{4}{x^4})(x^3 - \frac{4}{x^3})$.

1. -15

2. 15

3. -10

4. 10

If $a - b = 5$ and $ab = 25$, then the value of $a^3 - b^3$ is:

- 1. 525
- 2. 500
- 3. 450
- 4. 400

If $x - y = 25$ and $xy = 444$, compute the value of $x^3 - y^3$.

- 1. 48,925
 - 2. 42,985
 - 3. 28,495
 - 4. 26,725
-

If $a = (\sqrt{2} - 1)^{\frac{1}{3}}$, then the value of $\left(a - \frac{1}{a}\right)^3 + 3\left(a - \frac{1}{a}\right)$ is:

2

-2

$\sqrt{2}$

$-\sqrt{2}$

If $x^3 = 184 + y^3$ and $x = 4 + y$, then the value of $(x + y)$ is (given that $x > 0$ and $y > 0$):

1. $2\sqrt{14}$

2. $2\sqrt{11}$

3. $-2\sqrt{14}$

4. $-2\sqrt{11}$

If for a non-zero x , $7x^2 + 5x + 7 = 0$, then the $x^3 + \frac{1}{x^3}$ is _____.

1. $\frac{420}{343}$

2. $\frac{610}{420}$

3. $\frac{610}{343}$

4. $\frac{343}{610}$

If $\left(x - \frac{1}{x-1}\right) = 3$, then the value of $(x-1)^2 + \frac{1}{(x-1)^2}$ is:

1. 6

2. 4

3. 3

4. 5

If $x + y = 12$ and $xy = 6$, then the value of $\frac{x}{y} + \frac{y}{x}$ is equal to:

1. 18

2. 22

3. 24

4. 26

If $a-b=3$ and $a^3-b^3=999$,
then find the value of a^2-b^2 .

1.60

2.62

3.64

4.63

If $2x+y=8$, $xy=6$ find $4x^2+y^2$?

16

24

28

40

If $x^2/y^2 + y^2/x^2 = 7$, A possible value of $x^3/y^3 + y^3/x^3$?

16

15

17

18

If $r + 64/r = 16$, find $r^4 + 1/r^3$?

4096(1/512)

512

512(1/4096)

4096

If $a + b = 12$, $ab = 35$, $a^4 + b^4 - 10^3$?

2226

2026

2126

1226

If $2a+5b=12$, $ab=3$, find $4a^2+25b^2$?

44

64

24

84

If $2a+b=10$, $2ab=9$, $2a>b$, find $2a-b$?

6

8

10

12

If $3x+y = 12$, $xy = 9$, find $3x-y$?

4

5

6

7

If $(a+1/a+3)^2=16$, a is non zero real number, find a^2+1/a^2 ?

3

47

49

7

If $4x-3y=12$, $xy=5$, find $(16x^2+9y^2)/8$?

33

18

3

44

If $5x-1/4x = 6$, $x>0$, find $25x^2-1/16x^2$?

$6\sqrt{41}$

36

$\sqrt{246}$

$6\sqrt{31}$

If $4a-3b=1$, $ab=\frac{1}{2}$, $a>0$,
 $b>0$, find $64a^3+27b^3$?

15

25

30

35

If $a^3+b^3=218$, $a+b=2$, find
 $\sqrt{1-ab}$?

3

4

5

6

If $x - 1/x = 1$, find
 $x^8 + 1/x^8$?

119

-1

3

47

If $2x - 3/x = 2$, find
 $16x^4 + 81/x^4$?

328

180

184

220

If $4x^4 - 37x^2 + 9 = 0$, $x > \sqrt{3}/\sqrt{2}$,
find $8x^3 - 27/x^3$?

35

-215

215

-35

$X^5 + 1/X^5 = 55449$, find
 $x^3 + 1/x^3$?

702

110

321

2

If $x + \frac{1}{x} = 6$, find
 $x^5 + \frac{1}{x^5} - 6720$?

0

5

6

6726

$x^2 - 3x + 1 = 0$, find $(x^2 - 3)(x^2 - 4)$?

10

12

13

NONE

$x^2 - 3x + 1 = 0$, find $(x^3 - 9)^2$?

60

70

80

100

If $a - 12/a = 1$, $a > 0$, find $a^2 + 16/a^2$?

15

19

17

11

If $4x^4 = 5x^2 - 1$, $x > 1/\sqrt{2}$, Find $(2x^2 - x - 1)$?

0

1

2

-2

If $x - \frac{2}{x} = 15$, find $x^2 + \frac{4}{x^2}$?

229

227

221

223

If $6x^2 - 5x - 9 = 0$, find
 $36x^2 + 81/x^2$?

111

122

133

144

If $\sqrt{x} + 1/\sqrt{x} = \sqrt{6}$, find
 $x^6 + 1/x^6$?

2712

2270

2502

2702

If $x + 81/x = 18$, ($x > 0$),
find $x^2 + 162/x^2$?

78

81

83

85

If $x + 1/15x = 3$, find
 $9x^3 + 1/375x^3$?

237.6

273.6

367.2

376.2

If $5x + \frac{1}{3x} = 25$, find
 $27x^3 + \frac{1}{125x^3}$?

3347

3348

3345

3346

If $2x - \frac{5}{6x} = 3$, find
 $\frac{32x^3}{5} - \frac{25}{54x^3}$?

35.8

33.6

32.6

24.6

If $a - \frac{24}{a} = 5$, ($a > 0$), find $a^2 + \frac{64}{a^2}$?

45

56

60

65

If/यदि $x^8 + \frac{1}{x^8} = 14$, then/तो $x^3 + \frac{1}{x^3} = ?$

A. $\sqrt{2 - \sqrt{6}} (\sqrt{6} - 1)$

B. $\sqrt{2 + \sqrt{6}} (\sqrt{6} - 1)$

C. $\sqrt{2 + \sqrt{6}} (\sqrt{6} + 1)$

D. $\sqrt{2 - \sqrt{6}} (\sqrt{6} + 1)$

If $a^4 + \frac{1}{a^4} = 50$, ($x > 0$), find $a^3 + \frac{1}{a^3}$.

$$\sqrt{2(1 + \sqrt{13})}(-1 - 2\sqrt{13})$$

$$\sqrt{2(1 - \sqrt{13})}(-1 + 2\sqrt{13})$$

$$\sqrt{2(1 + \sqrt{13})}(-1 + 2\sqrt{13})$$

$$\sqrt{2(1 - \sqrt{13})}(-1 - 2\sqrt{13})$$

If $8k^6 + 15k^3 - 2 = 0$, Find $k + \frac{1}{k}$?

$$2\left(\frac{1}{2}\right)$$

$$2\left(\frac{1}{8}\right)$$

$$8\left(\frac{1}{2}\right)$$

$$8\left(\frac{1}{8}\right)$$

**If $27k^6 + 80k^3 - 3 = 0$
then the positive value
of $k + 1/k$**

**If $8k^6 - 31k^3 - 4 = 0$
Find negative value of
 $K + 1/K$**

If $x^2 - 4x + 1 = 0$, find $x^9 + x^7 - 194x^5 - 194x^3$.

- a. 4
- b. $2\sqrt{3}$
- c. -4
- d. $-2\sqrt{3}$

If $x^2 - 9x + 1 = 0$, what is the value of $x^8 - 6239x^4 + 1$?

- 1.1**
- 2.0**
- 3.-1**
- 4.2**

If $x^2 - 8x + 1 = 0$, find
 $x^8 - 3842x^4 + 1$?

2

0

1

-1

$x^2 - 3x + 1 = 0$, find $x^{12} + x^8 - 123x^7 - 123x^3$.

a. 7

b. -7

c. $\sqrt{13}$

d. $-\sqrt{13}$

If $x^2 - 6x + 1 = 0$,

Find $x^9 - 1153x^5 - 198x^2 + 2x$.

a. $\sqrt{32}$

b. $-\sqrt{32}$

c. 4

d. -4

If $x = \frac{5+\sqrt{21}}{2}$, find $x^{11} + x^7 - 2525x^6 - 527x^3$.

a. 5

b. -5

c. $\sqrt{5}$

d. $-\sqrt{5}$

$$(x + y)(x - y)(x^2 + y^2)(x^4 + y^4)(x^8 + y^8)(x^{16} + y^{16})$$

$$x^{128} - y^{128}$$

$$x^{64} - y^{64}$$

$$x^{256} - y^{28}$$

$$x^{32} - y^{32}$$

What is the value of $(k - \frac{1}{k})(k^2 + \frac{1}{k^2})(k^4 + \frac{1}{k^4})(k^8 + \frac{1}{k^8})(k^{16} + \frac{1}{k^{16}})(k^{32} + \frac{1}{k^{32}})$

$(k - \frac{1}{k})(k^2 + \frac{1}{k^2})(k^4 + \frac{1}{k^4})(k^8 + \frac{1}{k^8})(k^{16} + \frac{1}{k^{16}})(k^{32} + \frac{1}{k^{32}})$ का मान है।

a) $\frac{(k^{64} - \frac{1}{k^{64}})}{(k + \frac{1}{k})}$

b) $\frac{(k^{32} - \frac{1}{k^{32}})}{(k - \frac{1}{k})}$

c) $\frac{(k^{32} + \frac{1}{k^{32}})}{(k + \frac{1}{k})}$

d) $\frac{(k^{32} + \frac{1}{k^{32}})}{(k + \frac{1}{k})}$

$$x - y = \sqrt{5}$$

$$10 + xy = 13/xy$$

Find $x^4 + y^4$?

50

51

52

53

$$x - y = \sqrt{3}$$

$$6 + xy = 6/xy$$

Find $x^4 + y^4$?

20

21

22

23

Thankyou

Algebra

Class -3

If $\frac{7(x^2+1)-17x}{7x} = 2$, then $\left(\frac{x+1}{\sqrt{x}}\right)^2 = ?$

यदि $\frac{7(x^2+1)-17x}{7x} = 2$ है तो, $\left(\frac{x+1}{\sqrt{x}}\right)^2 = ?$

A. 6.42

B. 6.53

C. 6.86

D. 6.31

If/यदि $(a^2 - 16) = -\frac{2}{a^2}$, then/तो $\sqrt{\frac{156a^2 \times 250}{13a^4 + 104a^2 + 26}} = ?$

- A. $5\sqrt{5}$
- B. $6\sqrt{5}$
- C. $7\sqrt{5}$
- D. $10\sqrt{5}$

If $(37x^2 - 43x + 43)(10x^2 - 8x + \frac{82}{7})^{-1} = 3.5$ then $\sqrt{x^2 + \frac{1}{x^2} + 10.75} = ?$

- 81
- 65
- 70
- NONE.

If $x^2+x=43$, then $(x+7)^2+1/(x+7)^2$?

78

79

81

83

If $x^2 + x = 19$, find $(x + 5)^2 + \frac{1}{(x+5)^2}$.

a. 77

b. 79

c. 81

d. 83

Expand x^2+2x+3 , About $x = -2$?

1. $(x - 2)^2 - 2(x + 2) + 3$

2. $(x + 2)^2 - 2(x + 2) + 3$

3. $(x + 2)^2 + 2(x + 2) + 3$

4. $(x - 2)^2 - 2(x - 2) - 3$

If/यदि $x^2 - 3\sqrt{7}x + 13 = 0$, then/तो $(x - \sqrt{7})^6 - \frac{1}{(x - \sqrt{7})^6} = ?$

A. $80\sqrt{77}$

B. $-80\sqrt{77}$

C. $\sqrt{77}$

D. *none*

If $x^2 - 38x + 96 = 0$, find $(x - 5) - \frac{69}{x-5}$.

27

28

29

30

$$x^3 - 4x^2 + 19 = 6(x - 1)$$

Find $[x^2 + \{1/(x-4)\}]$?

2

4

6

8

If $x^3 - 6x^2 + 35 = 8(x - 2)$, find $x^2 + \frac{3}{x-6}$.

2

4

6

8

If $(x+5)(x+6) + \{9/x(x-1)\} = 0$,
Find $3x^2 + 15x - 1$?

4

6

8

10

Find $(x+1)(x+2)+\{1/x(x-1)\} = 0$

Find $x^4+x^2+2x^3 -1$?

0
1
2
-1

Special Cases

1) $X + 1/X = 1$

2) $X + 1/X = -1$

3) $X + 1/X = \sqrt{3}$

4) $X + 1/X = \sqrt{2}$

If $\sqrt{x} + 1/\sqrt{x} = 1$, then find

a) $X^3 - 1$

b) $x^{40} + 1/x^{40}$

If $x + 1/x = 1$, Find

a) $x^{44} + x^{41} + x^{47} + x^{38}$

b) $x^{94} + 1/x^{94}$

If $x + 1/x = \sqrt{3}$, find

a) $x^{41} + x^{42} + x^{43} + x^{47} + x^{49} + x^{48}$?

b) $x^{100} + 1/x^{100}$

c) $x^{33} + 1/x^{33}$

If/यदि $\sqrt{3}x + 1 + \frac{1}{\sqrt{3}x} = 0$, then/तो $\frac{3\sqrt{3}x^3}{\sqrt{7}} = ?$

A. $2\sqrt{7}$

B. 1

C. $\sqrt{7}$

D. $\frac{1}{\sqrt{7}}$

If $\frac{x}{3\sqrt{7}} + 1 = \frac{-3\sqrt{7}}{x}$, then $\sqrt{x^{121} - 189\sqrt{7}x^{118} + 1728} = ?$

यदि $\frac{x}{3\sqrt{7}} + 1 = \frac{-3\sqrt{7}}{x}$, तो $\sqrt{x^{121} - 189\sqrt{7}x^{118} + 1728} = ?$

24

24√2

24√3

144

If $3\sqrt{3}/x^2 + 3\sqrt{3} = 9/x$

then

$\sqrt{(x^{144} + 48x^{12} + 95)}$?

12

12√2

12√3

24

If $7(2\sqrt{2} - x)/x = 7 + 14/x^2$

Then

$\sqrt{23x^{64} + 42x^{12} + 163}$

12

$12\sqrt{2}$

$12\sqrt{3}$

24

PANACEA METHOD VS MANIPULATION

If $x^2+2=2x$, then $x^4-x^3+x^2+2$?

- a. 0
- b. 1
- C. -1
- d. $\sqrt{2}$

If $x^2+2 = 2x$, find $x^4-3x^3+3x^2-2$?

- A. $x+1$
- B. x
- C. $2x$
- D. 0

If $x^2 - 1 = -3x$, find $x^4 + 2x^3 - 3x^2 + 4x - 1$?

- A. 0
- B. $3x$
- C. x^2
- D. $x - 1$

If $x^3 + x = 1$, find $x^5 + x^4 + x - 2$?

- A. 0
- B. x
- C. $-3x^3$
- D. $-2x^2$

If $x^3+4x=8$, find x^7+64x^2 ?

32

64

128

256

G.P IN ALGEBRA

If $a^3 + 3a^2 + 9a = 1$, Find $a^3 + \frac{3}{a}$

26

27

28

29

If $x^3 + 7x^2 + 49x - 1 = 0$, Find $x^3 + \frac{7}{x}$

343

344

345

346

If $a^4 + 3a^3 + 9a^2 + 27a = 1$, find $a^4 + \frac{3}{a}$.

80

81

82

83

If $x^5 + x^4 + x^3 + x^2 + x = 0$, Find $x^{1000} + x^{100} + x^{10}$.

0

1

2

3

If $a+a^2+a^3+a^4-2=0$, Find a^4+2/a ?

- 1
- 2
- 3
- 4

If $x^2-5x+25=0$ then X^5-5x^4-3100 ?

- 25
- $25\sqrt{2}$
- $25\sqrt{3}$
- None.

If $\frac{x}{3} + 1 = -\frac{3}{x}$, find $x^5 - 26x^2 + 3x + 14$.

2

3

4

5

28.If $\frac{1}{7} + \frac{1}{x} = \frac{1}{x+7}$, then $x^{49} - 343x^{46} + x^3 + 17$.

a. 324

b. 0

c. 360

d. 441

Concept

If $ax + by = m$,
 $ay - bx = n$

$a^2 + b^2 = 10$, $x^2 + y^2 = 17$, $ax + by = 7$,
find $ay - bx = ??$

- A. 9
- B. 10
- C. 11
- D. 12

If $ax+by=8$, $ay-bx = 6$, $a^2+b^2+x^2+y^2 = 29$

Find $(a^2+b^2)/(x^2+y^2)$; $a^2+b^2 > x^2+y^2$?

25/4

27/4

23/4

29/4

Thankyou

Algebra

Class -4

**$X = 1 + 2\frac{2}{3} + 2\frac{1}{3}$; find
 $x^3 - 3x^2 - 3x - 1$?**

0

-1

-2

-3

If $y = 1 + \sqrt{3} + \sqrt{2}$, find
 $y^4 - 4y^3 - 4y^2 + 16y - 8$?

- A) 14
- B) -14
- C) 0
- d) 7

If $y = 1 + \sqrt{3} + \sqrt{2}$, find
 $2y^4 - 8y^3 - 5y^2 + 26y - 28$?

- A) $4\sqrt{6}$
- B) $6\sqrt{6}$
- C) $3\sqrt{6}$
- d) $2\sqrt{6}$

If $y = 1 + \sqrt{3 + \sqrt{4}}$, find
 $2y^4 - 8y^3 - 6y^2 + 28y - 84$?

- A) $40\sqrt{3}$
- B) $-40\sqrt{3}$
- C) $24\sqrt{3}$
- d) $36\sqrt{3}$

If $x^2 + 7 = 3x$, Then
 $2x^4 + 6x^2 - 3x^3 + 18x + 7$?

- A. 1
- B. -1
- C. 0
- D. 2

If $a + \frac{1}{b} = 1$ and $b + \frac{1}{c} = 1$,

then $c + \frac{1}{a}$ is equal to

- (1) 0 (2) $\frac{1}{2}$
 (3) 1 (4) 2

(SSC CGL Prelim Exam. 04.02.2007
 (First Sitting))

If $\frac{x}{2} + \frac{2}{y} = 1$ and $\frac{y}{2} + \frac{2}{z} = 1$, then $\frac{z}{2} + \frac{2}{x} = ?$

यदि $\frac{x}{2} + \frac{2}{y} = 1$ और $\frac{y}{2} + \frac{2}{z} = 1$ हो, तो $\frac{z}{2} + \frac{2}{x} = ?$

1. 0
 2. 2
 3. 1
 4. -1

If $x + \frac{1}{x} = a$, $x - \frac{1}{x} = b$, $y + \frac{1}{y} = m$ and $y - \frac{1}{y} = n$?

A. $xy + \frac{1}{xy}$?

B. $\frac{x}{y} + \frac{y}{x}$?

$X + \frac{1}{X} = 7$, $Y - \frac{1}{Y} = 4$,
Find $XY + \frac{1}{XY}$?

12√5

13√5

7√5

11√5

If $x^2 + 1/x^2 = 34$, $y^2 + 1/y^2 = 23$,

Find $XY + 1/XY$?

$a - 1/b = m$, $b - 1/c = n$, $c - 1/a = p$

What is $abc - 1/abc$?

$$a + \frac{1}{b} = m, b + \frac{1}{c} = n, c + \frac{1}{a} = p$$

What is $abc + \frac{1}{abc}$?

If/यादि $a + \frac{1}{b} = 5, b + \frac{1}{c} = 3, c + \frac{1}{a} = \frac{7}{5}$ then/तो $abc = ?$

A. $\frac{29 - 4\sqrt{51}}{5}$

B. $\frac{29 + 4\sqrt{51}}{5}$

C. $\frac{29 + \sqrt{51}}{5}$

D. $\frac{29 - \sqrt{51}}{5}$

If $xyz = 1$, $x + \frac{1}{z} = 5$, $y + \frac{1}{x} = 29$, find $\left(z + \frac{1}{y}\right)$.

- a. $\frac{1}{3}$
- b. $\frac{1}{4}$
- c. $\frac{1}{6}$
- d. 4

$a + \frac{1}{b} = 1$, $b + \frac{1}{c} = 1$, find abc ?

1

-1

BOTH -1 & 1

0

If $a + \frac{1}{b} = b + \frac{1}{c} = c + \frac{1}{a}$,
then what is the value of ABC?

1

-1

BOTH -1 & 1

0

$a + \frac{1}{b} = \frac{7}{3}$, $b + \frac{1}{c} = \frac{13}{4}$, $c + \frac{1}{a} = \frac{9}{2}$

Find abc ?

21

22

23

24

If $a - \frac{1}{a} = b$, $b - \frac{1}{b} = c$, $c - \frac{1}{c} = a$, find $\frac{a^2b^2+b^2c^2+c^2a^2}{(abc)^2}$.

.If $a - \frac{1}{a} = b$, $b - \frac{1}{b} = c$, $c - \frac{1}{c} = a$, find $\left(\frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca}\right)$.

IRRELEVANT

If $a - \frac{1}{a} = b$, $b - \frac{1}{b} = c$, $c - \frac{1}{c} = a$, find abc .

Concept:

$$(ax^2+by^2+cxy)(ax^2+by^2-cxy)$$

$$a^2x^4 + (2ab-c^2)x^2y^2 + b^2y^4$$

If $1 + x^2 + x^4 = 585$ and $1 - x + x^2 = 65$, then find the value of $1 + x + x^2$.

- 1.8
- 2.7
- 3.6
- 4.9

$x^4 + x^2y^2 + y^4 = 24$, $x^2 + xy + y^2 = 6$, $x^3 - y^3$?

- 3
- 4
- 5
- none

$$a+ar+ar^2 = 42$$

$$a^2+a^2r^2+a^2r^4 = 1092$$

Find Product of a & r ?

4

6

8

10

If $x^4+y^4+x^2y^2=117$, $x^2+y^2-xy=3(4+\sqrt{3})$, find x^2+y^2 ?

$6\sqrt{3}$

$13\sqrt{3}$

9

12

If $x^4+y^4+x^2y^2=21$, $x^2+y^2+xy=3$, find $(-xy)$?

-2

2

-1

1

$$5x^2+3xy+7y^2 = 19,$$

$$5x^2-3xy+7y^2 = 6$$

find $25x^4+61x^2y^2+49y^4$?

110

112

114

116

$$25x^4 + 61x^2y^2 + 49y^4 = 114$$

$$5x^2 + 3xy + 7y^2 = 19$$

find $5x/y + 7y/x$

25/13

75/13

150/13

Wrong Question.

If/यदि $x^4 + x^2y^2 + y^4 = 222$, $x^2 + xy + y^2 = 37$ then/तो $(x + y) = ?$

A. $-\sqrt{\frac{105}{2}}$

B. $\pm\sqrt{\frac{105}{2}}$

C. $\pm\sqrt{105}$

D. $\sqrt{\frac{105}{2}}$

If/यदि $x^4 + x^2y^2 + y^4 = 80, x^2 + xy + y^2 = 10$, then/तो $(x^3 - y^3) = ?$

A. $11\sqrt{7}$

B. $9\sqrt{7}$

C. $10\sqrt{7}$

D. $\sqrt{7}$

If/यदि $\frac{(117)^4 + (117)^2 + 1}{(117)^3 + 1} = p - \frac{q}{r}$, then/तो $(q - p + r) = ?$

A. 0

B. 1

C. -1

D. -2

$$\text{If } \frac{a^4 + a^2 + 1}{a^3 + 1} = p + \frac{q}{r}, \text{ find } (p + q + r).$$

- a. $(a + 1)$
- b. $2a + 1$
- c. $2(a + 1)$
- d. $a + 2$

$$3x^2 + 8xy + 16y^2 = 5$$

$$3x^2 - 8xy + 16y^2 = 7$$

Find

$$9x^4 + 256y^4 + 32xy = ?$$

- 35
- 40
- 45
- 30

$$9x^4 + 38xy + 81y^4 = 42$$

$$3x^2 + 4xy + 9y^2 = 7$$

Find $3x/y + 9y/x$?

48

50

52

56

$$\text{If } (a - b) = m, \quad (a + b) = n, \quad a^3 - b^3 = x,$$

$$a^3 + b^3 = y, \quad \text{find } (a^4 + a^2b^2 + b^4).$$

- a. $\frac{xm}{yn}$
- b. $\frac{xy}{mn}$
- c. $\frac{mn}{xy}$
- d. *NOT*

If $A^3 - B^3 = 12.5$, $A^3 + B^3 = 16$, find $A^4 + A^2B^2 + B^4 \times A^2 - B^2$

यदि $A^3 - B^3 = 12.5$, $A^3 + B^3 = 16$ हो तो, $(A^4 + A^2B^2 + B^4) \times (A^2 - B^2)$

- A. 125
- B. 150
- C. 175
- D. 200

If $x+y=13$, find $(x-8)^3 + (y-5)^3$?

- 0
- 13
- 169
- 2197

If $x+y=25$, find $(x-16)^3+(y-10)^3$?

0

125

625

225

If $(a-1)^2+(b+2)^2+(c-1)^2=0$, find $a^2+b^2+c^2$?

0

1

2

6

If $a^2+b^2+49c^2+18=2(b-28c-a)$, find $a+b-7c$?

- 1
- 2
- 3
- 4

If $(a+b-c-8)^2+(2b+2c-a-9)^2+(c+2a-b-11)^2=0$, find $a+b$?

- 11
- 11
- 14
- 14

if $a^2+b^2+c^2+216=12(a+b-2c)$, find $\sqrt{ab-bc-ca}$?

6 $\sqrt{5}$

4 $\sqrt{5}$

8 $\sqrt{5}$

3 $\sqrt{5}$

If $a^2+4b^2+c^2+70 = 2(3a-10b+6c)$

Then Find $2ab + 3c$

2

3

-3

-2

$x^2+4y^2+3z^2+19/4 = 2\sqrt{3}(x+y+z)$, find $(x-4y+3z)$?

$2\sqrt{3}$

$-\sqrt{3}$

$\sqrt{3}$

$-2\sqrt{3}$

If $121a^2+2b^2+25c^2=66a+2\sqrt{2}b-30c-19$,
find $(22a)^2-(\sqrt{2}b)^3+5c$?

38

36

34

32

If $2x^2+4y^2-4xy+14x+49=0$, find $(8y+5x)/2xy$?

9/7

-9/7

2/7

-2/7

If $2x^2+y^2+2xy-2\sqrt{5}x+5=0$,
find $x^3-5x^2+x^2y^2$?

5

$\sqrt{5}$

$5\sqrt{5}$

$-5\sqrt{5}$

If $5x^2+9y^2+12xy = -8(2+x)$

Find $3y-x$

10

12

14

16

If $a^2+b^2+49c^2+18=2(b-28c-a)$,
find $a+b-7c$?

1

2

3

4

if $a^2+b^2+c^2+216=12(a+b-2c)$, find $\sqrt{ab-bc-ca}$?

6 $\sqrt{5}$

4 $\sqrt{5}$

8 $\sqrt{5}$

3 $\sqrt{5}$

If $a^2+4b^2+c^2+70 = 2(3a-10b+6c)$

Then Find $2ab + 3c$

2

3

-3

-2

$x^2+4y^2+3z^2+19/4 = 2\sqrt{3}(x+y+z)$, find
 $(x-4y+3z)$?

$2\sqrt{3}$

$-\sqrt{3}$

$\sqrt{3}$

$-2\sqrt{3}$

If $121a^2+2b^2+25c^2=66a+2\sqrt{2}b-30c-19$,
find $(22a)^2-(\sqrt{2}b)^3+5c$?

38

36

34

32

If $2x^2+4y^2-4xy+14x+49=0$, find $(8y+5x)/2xy$?

9/7

-9/7

2/7

-2/7

If $2x^2+y^2+2xy-2\sqrt{5}x+5=0$,
find $x^3-5x^2+x^2y^2$?

5

$\sqrt{5}$

$5\sqrt{5}$

$-5\sqrt{5}$

If $5x^2+9y^2+12xy = -8(2+x)$

Find $3y-x$

10

12

14

16

If/यदि $x^8 + 25 = \frac{x^4}{y^2} (24y^2 - y^4 - 49)$, then/तो $\frac{2x^4 - y^2}{x^4 y^2} = ?$

A. 1

B. $-3/35$

C. $3/35$

D. none

If/यदि $x^8 + 7 = \frac{x^4}{y^2} \left((14 + 2\sqrt{7})y^2 - y^4 - 49 \right)$, then/तो $\sqrt{7}x^4y^2 = ?$

A. $\pm \frac{1}{\sqrt{7}}$

B. 49

C. -49

D. $\frac{1}{\sqrt{2}}$

7

$$\text{If } (a^2+1)(b^2+1) + N^2 = 2N(a+b) \\ \text{Then } a + \frac{1}{a} = N$$

If/यदि $(a^2 + 1)(b^2 + 1) + 5 = 2\sqrt{5}(a + b)$, then /तो $a^3 + \frac{1}{a^3} = ?$

- A. $5\sqrt{2}$
- B. $2\sqrt{2}$
- C. $2\sqrt{5}$
- D. $2\sqrt{2} + \sqrt{3}$

If/यदि $(a^2 + 1)(b^2 + 1) + 8 + 4\sqrt{2}(a + b) = 0$ then/तो $a^5 + \frac{1}{a^5} = ?$

- A. $-\frac{49}{12a}$
- B. 58
- C. $58\sqrt{2}$
- D. $-58\sqrt{2}$

Thankyou

Algebra

Class -5

If $6\sqrt{6} p^3 + 2\sqrt{2} q^3 = (\sqrt{6} p + \sqrt{2} q)(Sp^2 + Mq^2 - Npq)$, then the positive value of $\sqrt{S^2 + M^2 + 2N^2}$ is:

यदि $6\sqrt{6} p^3 + 2\sqrt{2} q^3 = (\sqrt{6} p + \sqrt{2} q)(Sp^2 + Mq^2 - Npq)$ है, तो $\sqrt{S^2 + M^2 + 2N^2}$ का धनात्मक मान क्या है?

- (a) 10
- (b) 8
- (c) 9
- (d) 12

If $54\sqrt{2}x^3 + 56\sqrt{7}y^3 = (3\sqrt{2}x + 2\sqrt{7}y)(Ax^2 - Bxy + Cy^2)$ find $A - B^2 + C$?

0

1

-1

458

$3\sqrt{3}(x+y)^3 - 2\sqrt{2}(x-y)^3 = (x\sqrt{3} - \sqrt{2}) + y(\sqrt{3} + \sqrt{2})(Ax^2 + By^2 + Cxy)$,
find $(A+B-C)^2$?

32

64

128

256

If $343(x+y)^3 + 216(x-y)^3 = (13x+y)(Ax^2 + By^2 - 2Cxy)$,
find $A+B+C$?

157

147

137

167

If $x^6 - 6\sqrt{6}y^6 = (x^2 + Ay^2)(x^4 + Bx^2y^2 + Cy^4)$, Find $(A^2 - B^2 + C^2)$?

32

34

36

38

$125x^6 + 7\sqrt{7}y^6 = (5x^2 + By^2)(25x^4 - ABx^2y^2 + 7y^4)$,
find $(2A - B)^2$?

107- $\sqrt{7}$

107-20 $\sqrt{7}$

107-4 $\sqrt{7}$

0

$27(x+y)^3 - 8(x-y)^3 = (x+5y)(Ax^2 + By^2 + Cxy)$,
Find $(A+B-C)$?

13

16

18

11

$(x+y)^3+8(x-y)^3=(3x+Ay)(3x^2+Bxy+Cy^2)$,
Find $A+B+C$?

- 0
- 2
- 3
- 4

If $8(x+y)^3-27(x-y)^3=(5y-x)(Ax^2+By^2+Cxy)$, find $A+B-C$?

- 32
- 34
- 36
- 38

If $(2x+y)^3 - (x-2y)^3 = (x+3y)(Ax^2+Cxy+By^2)$

Find $A+2B+C$?

7

10

12

16

If $8x^3 - (3x-2y)^3 = (2y-Ax)(Bx^2+4y^2-cxy)$

Find $2A+B-C$?

2

3

4

5

If $(x+2y)^3 - (2x-y)^3 = (Ay-x)(Bx^2+Cy^2+Dxy)$

Find A+B-C-D

- 0
- 1
- 4
- 2

If $(4x+2y)^3 + (4x-2y)^3 = 16(Ax^3+Bxy^2)$, then what is the value of $\frac{1}{2}(\sqrt{A^2+B^2})$?

यदि $(4x+2y)^3 + (4x-2y)^3 = 16(Ax^3+Bxy^2)$, तो $\frac{1}{2}(\sqrt{A^2+B^2})$ का मान क्या है?

- 3
- 5
- 7
- 8

If $(2x+3y)^3 - (2x-3y)^3 = 3y[Ax^2+By^2]$, then what is the value of $(A-B)$?

यदि $(2x+3y)^3 - (2x-3y)^3 = 3y[Ax^2+By^2]$ है, तो $(A-B)$ का मान क्या होगा?

3

6

9

12

Find the ratio of coefficient of x^3 to the coefficient of x in $(2x+3)^5$?

$(2x + 3)^5$ के विस्तार में x^3 के गुणांक और x के गुणांक का अनुपात ज्ञात कीजिए।

A 8:9

B 5: 12

C 4:7

D 6:11

If n be a positive integer. If the coefficient of 2nd, 3rd, 4th terms in the expansion of $(1+x)^n$ are in A.P. Then the value of n is:

यदि n एक धनात्मक पूर्णांक हो। यदि $(1+x)^n$ के विस्तार में दूसरे, तीसरे, चौथे पदों का गुणांक A.P में है तो n का मान है:

- 1. 9
- 2. 5
- 3. 8
- 4. 7

Evaluate $x^4 - 4x^3y + 6x^2y^2 - 4xy^3 + 2y^4$?

- A. $(x-y)^4 + y^4$
- B. $(x+y)^4 + y^4$
- C. $(x+y)^4 - y^4$
- D. $(x-y)^4 - y^4$

Evaluate $x^4+4x^3y+6x^2y^2+4xy^3-y^4$?

A. $(x-y)^4+2y^4$

B. $(x+y)^4+2y^4$

C. $(x+y)^4-2y^4$

D. $(x-y)^4-y^4$

Evaluate $2x^4-4x^3y+6x^2y^2-4xy^3+2y^4$?

A. $y^4+(x+y)^4+x^4$

B. $y^4+(x-y)^4-x^4$

C. $x^4-(x+y)^4-y^4$

D. $y^4+(x-y)^4+x^4$

$x = \sqrt{17} - 1$, find $17x^2 + 34x + 26$?

0

297

298

$\sqrt{298}$

If $x = \sqrt{2} - 1$, find

$3x^5 + 15x^4 + 30x^3 + 30x^2 + 15x + 5$?

$4 + 6\sqrt{2}$

$16 + 2\sqrt{2}$

$12\sqrt{2} + 8$

$2 + 12\sqrt{2}$

If $x=2\sqrt{2}-1$, find $19x^3+57x^2+57x+19$?

300 $\sqrt{2}$

302 $\sqrt{2}$

304 $\sqrt{2}$

306 $\sqrt{2}$

If $3(a^2 + b^2 + c^2) = (a + b + c)^2$,

which of the following values of a, b, c are possible?

a. $a = 1.25, b = 1, c = 0.5$

b. $a = 1.25, b = 1, c = 1.25$

c. $a = 1.25, b = 1.25, c = 1.25$

d. $a = 1.25, b = 1, c = 1$

If $x^2 + y^2 + z^2 = 3$ then which of the following cannot be a value of $(xy + yz + zx)$?

- 0
- 1
- 1
- 2

If $a = x$, $b = x + d$, $c = x + 2d$

Find $a^2 + b^2 + c^2 - ab - bc - ca$?

Find $a^3 + b^3 + c^3 - 3abc$?

If/यदि $x = a - \sqrt{3}, y = a, z = a + \sqrt{3}$, then/तो $(x^3 + y^3 + z^3 - 3xyz) = ?$

A. $37a$

B. a

C. $18a$

D. $27a$

4.

If $a = 2022$, $b = 2021$ and $c = 2020$, then value of $a^2 + b^2 + c^2 - ab - bc - ca$ is:

यदि $a = 2022$, $b = 2021$ और $c = 2020$ है, तो $a^2 + b^2 + c^2 - ab - bc - ca$ का मान ज्ञात कीजिए।

1

2

3

4

If $x = a, y = a + d, z = a + 2d,$

find $x^3 + y^3 + z^3 - 3xyz.$

- a. $(x + y + z)$
- b. $3(x + y + z)d^2$
- c. $(x + y + z)d^2$
- d. $3(x + y + z)d$

5. *If $x + y + z = a, xy + yz + zx = b, 3xyz = c,$ find $x^3 + y^3 + z^3.$*

- a. $a^3 - 3ab + c$
 - b. $a^3 - 3a^2b^2 + c^3$
 - c. $a^3 + 3ab - c$
 - d. $a^3 + 3ab + c$
-

If $x = (b - c)(a - d)$, $y = (c - a)(b - d)$, $z = (a - b)(c - d)$, then $\frac{(x+y)(y+z)(z+x)(x^3+y^3+z^3)}{9x^2y^2z^2} = ?$

- A. $\frac{1}{xyz}$
- B. 1
- C. $\frac{1}{3}$
- D. $\frac{-1}{3}$



$$x = a(b-c)$$

$$y = b(c-a)$$

$$z = c(a-b)$$

then find $\{(x+y)(y+z)(z+x)\}/4xyz$?

$$1/4$$

$$-1/4$$

$$-1/2$$

$$1/2$$

If $(2x+3)^3 + (x-8)^3 + (x+13)^3 = (2x+3)(3x-24)(x+13)$,
find the sum of all possible values of x ?

If $x = 32$, $y = 33$ and $z = 35$, then evaluate the expression $x^3 + y^3 + z^3 - 3xyz$.

यदि $x = 32$, $y = 33$ और $z = 35$ है, तो व्यंजक $x^3 + y^3 + z^3 - 3xyz$ का मान ज्ञात कीजिए।

700

900

1000

1120

If $\left(x + \frac{1}{2}\right)^3 + \left(3x + \frac{2}{3}\right)^3 + \left(x - \frac{1}{6}\right)^3 = \frac{1}{12} (9x+2)(2x+1)(6x-1)$, then what is the value of x?

यदि $\left(x + \frac{1}{2}\right)^3 + \left(3x + \frac{2}{3}\right)^3 + \left(x - \frac{1}{6}\right)^3 = \frac{1}{12} (9x+2)(2x+1)(6x-1)$ है, तो x का मान क्या होगा?

-1/5

-1/2

-1/4

-1/3

If $(7x-3)^3 + (x+2)^3 + 27(2x+5)^3 = 9(7x-3)(x+2)(2x+5)$, then $(9x+13):(5x+7)?$

11:6

2:1

5:3

35:19

If $(x+4)^3 + (2x+1)^3 + (2x+5)^3 = (6x+3)(x+4)(2x+5)$, then x ?

- 1
- 1
- 2
- 2

If $8x^3 + 27y^3 + 64z^3 = 72xyz$, find the relationship between x, y and z ?

- A. $2x+3y= 4z$
- B. $2x+3y= -4z$
- C. $2x+y+z=0$
- D. $2x-3y+4z=0$

If $x^2+y^2+z^2= 198x+202y+200z-30002$, find $x^3+y^3+z^3-3xyz$?

600

650

700

750

$x + y + z = 2$ and $xy + yz + zx = - 11$, then the value of $x^3 + y^3 + z^3 - 3xyz$ is:

A. 78

B. 71

C. 74

D. 69

$x + y + z = 7$ and $xy + yz + zx = 8$, then the value of $x^3 + y^3 + z^3 - 3xyz$ is:

- A. 200
- B. 150
- C. 125
- D. 175

$x + y + z = 11$ and $xy + yz + zx = 28$, then the value of $x^3 + y^3 + z^3 - 3xyz$ is:

- A. 1639
- B. 407
- C. 2255
- D. 1093

$x+y+z=17$, $xyz=171$ and $xy+yz+zx=111$
Find $(x^3+y^3+z^3+xyz)^{\frac{1}{3}}$

4

-4

2

-2

Thankyou

Algebra

Class -6

If $x + y + z = 7$, $x^2 + y^2 + z^2 = 85$
and $x^3 + y^3 + z^3 = 913$, then the value of $\sqrt[3]{xyz}$ is:

- 1
- 2
- 4
- 8

If $x + y + z = 1$ and $xy + yz + zx = xyz = -4$, then what is the value of $x^3 + y^2 + z^3$?

- A. 8
- B. -8
- C. 1
- D. -1

If $x+y+z= 3$, $xy + yz + zx=-12$ and $xyz = -16$, then the value of $\sqrt{x^3+y^3 + z^3+13}$ is:

- A. 9
- B. 8
- C. 10
- D. 11

If $x + y + z = 6$, $x^2 + y^2 + z^2 = 32$
and $x^3 + y^3 + z^3 = 189$, then the value of $4xyz$ is:

16

12

9

8

If $x + y + z = 2$, $x^3 + y^3 + z^3 - 3xyz = 74$, then
($x^2 + y^2 + z^2$) is equal to:

12

18

26

32

If $a+b+c=7$, $a^3 + b^3 + c^3 - 3abc = 301$, $ab + bc + ac = ?$

- A. 3
- B. 2
- C. -4
- D. -2

If $a + b + c = 4$, $ab + bc + ac = -14$, $abc = -18$, then
 $\sqrt{4a^3+4b^3+4c^3-36}$

- 24
- 26
- 28
- 30

If $2x + 3y + 4z = 11$, $8x^3 + 27y^3 + 64z^3 = 105$ & $xyz = 1$, then
 $4x^2 + 9y^2 + 16z^2 - 6xy - 12yz - 8xz = ?$

- 3
- 4
- 5
- 6

If $a+b+c=5$, $a^3 + b^3 + c^3 - 3abc = 185$, the value of $ab + bc + ac$
will be between:

- A. -7 & -3
- B. 1 & 5
- C. -3 & 1
- D. 5 & 9

If $a^3 + b^3 + c^3 - 3abc = 1.5$, $x = 2a + 2b$,

$y = 2b + 2c$, $z = 2c + 2a$,

find $x^3 + y^3 + z^3 - 3xyz$.

- a. 8
- b. 12
- c. 16
- d. 24

If $x^3 + y^3 + z^3 = \frac{3}{2}(1 + 2xyz)$,

$a = y + z - x$, $b = z + x - y$, $c = x + y - z$,

find $a^3 + b^3 + c^3 - 3abc$.

- a. 4
- b. 6
- c. 8
- d. 12

If $x^3 + y^3 + z^3 = \frac{3}{5}(2 + 5xyz)$

$a = 3x + 3y$, $b = 3y + 3z$, $c = 3z + 3x$

find $a^3 + b^3 + c^3 - 3abc$

32

64.8

64

60.8

If $x + y + z = 13$, $x^2 + y^2 + z^2 = 91$ & $xz = y^2$, then $z - x = ?$

A. 9

B. 8

C. 3

D. 5

if $x+y+z = -7$, $x^2+y^2+z^2 = 133$ & $xz = y^2$
then $z-x = ?$ ($x < z < y$)

- 4
- 5
- 6
- 7

$X+y+z=19$, $x^2+y^2+z^2=133$, $xz=y^2$,
find y ?

- 4
- 6
- 8
- cnd

If $a + b + c = 16$, $a^2 + b^2 + c^2 = 96$, $ab = c^2$. What is the value of $(b-a)$?

यदि $a + b + c = 16$, $a^2 + b^2 + c^2 = 96$, $ab = c^2$ है, तो $(b-a)$ का मान क्या है?

A. $\sqrt{21}$

B. $\sqrt{23}$

C. $1/3$

D. 3



If $(a + b + c) = 16$, and $(a^2 + b^2 + c^2) = 90$, find the value of $(ab + bc + ca)$.

यदि $(a + b + c) = 16$, और $(a^2 + b^2 + c^2) = 90$ है, तो $(ab + bc + ca)$ का मान ज्ञात कीजिये।

a) 84

b) 83

c) 82

d) 81

If $6^{x+y} = 216$ and $5^{x-y} = 3125$, then the value of $x + 2y$ is:

यदि $6^{x+y} = 216$ और $5^{x-y} = 3125$ है, तो $x + 2y$ का मान क्या है?

a) 4

b) 2

c) 3

d) 5

If $2^{x+2y} = 256$ and $3^{x-2y} = \frac{1}{27}$, then the value of $x + y$ is:

यदि $2^{x+2y} = 256$ और $3^{x-2y} = \frac{1}{27}$ है, तो $x + y$ का मान क्या है?

a) $\frac{19}{4}$

b) $\frac{21}{4}$

c) $\frac{23}{4}$

d) $\frac{25}{4}$

If $p = \frac{x^2+2x+1}{x^2-4x-5} \times \frac{x^4-8x}{x^3-x^2-2x} \div \frac{x^2+2x+4}{x^2-25}$, then what is the value of p , when $x = 2.5$?

यदि $p = \frac{x^2+2x+1}{x^2-4x-5} \times \frac{x^4-8x}{x^3-x^2-2x} \div \frac{x^2+2x+4}{x^2-25}$, तो p का मान क्या है, जब $x = 2.5$ है?

- | | |
|--------|--------|
| a) 2.5 | b) 7.5 |
| c) 1.5 | d) 5.5 |

If $x = 4.5$ and $y = 1.5$, then the value of

$\frac{x^4-y^4}{(x+y)^3-3xy(x+y)} \times \frac{x}{(x+y)^2-2xy} \div \frac{(x+y)^2-4xy}{(x+y)^2-3xy}$ is:

यदि $x = 4.5$ और $y = 1.5$ है, तो

$\frac{x^4-y^4}{(x+y)^3-3xy(x+y)} \times \frac{x}{(x+y)^2-2xy} \div \frac{(x+y)^2-4xy}{(x+y)^2-3xy}$

इसका मान क्या है:

- | | |
|------|--------|
| a) 6 | b) 2.5 |
| c) 3 | d) 1.5 |

If $x = 3.5$, then what is the value of

$$\frac{x^2+3x+9}{x^2-25} \div \frac{x^3-27}{x^2+3x-10} \times \frac{(x-4)^2-1}{x^2-4} ?$$

यदि $x = 3.5$ है, तो $\frac{x^2+3x+9}{x^2-25} \div \frac{x^3-27}{x^2+3x-10} \times \frac{(x-4)^2-1}{x^2-4}$ का मान क्या है? ?

a) $\frac{2}{9}$

b) $\frac{2}{11}$

c) $\frac{1}{4}$

d) $\frac{1}{5}$

If a, b, c are non- zero numbers and $14^a = 36^b = 84^c$, then $6b(\frac{1}{c} - \frac{1}{a})$ is equal to?

यदि a, b, c गैर-शून्यक संख्याये हैं और

$14^a = 36^b = 84^c$ हैं, तो $6b(\frac{1}{c} - \frac{1}{a})$ बराबर है

a)6

b)3

c)4

d)1.5

If $2^x = 3^y = 6^{-z}$, find the value of $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$.

0

1

$\frac{1}{2}$

$\frac{1}{4}$

If $3.7^x = 0.037^y = 10000$, then what is the value of $\frac{1}{x} - \frac{1}{y}$?

यदि $3.7^x = 0.037^y = 10000$, तो $\frac{1}{x} - \frac{1}{y}$ का मान क्या है?

(A) 1

(B) 2

(C) $\frac{1}{2}$

(D) $\frac{1}{4}$

Simplify the following expression:

निम्नलिखित व्यंजक का मान ज्ञात कीजिए:

$$\frac{(a^2 - 4b^2)^3 + 64(b^2 - 4c^2)^3 + (16c^2 - a^2)^3}{(a - 2b)^3 + (2b - 4c)^3 + (4c - a)^3}$$

A. $-(a+2b)(b+2c)(4c+a)$

B. $2(a+2b)(b+2c)(4c+a)$

C. $(a+2b)(b+2c)(4c+a)$

D $4(a+2b)(b+2c)(4c+a)$

Simplify the following expression.

$$\frac{5(a^6 - b^6)^3 + 5(b^6 - c^6)^3 + 5(c^6 - a^6)^3}{2(a^3 - b^3)^3 + 2(b^3 - c^3)^3 + 2(c^3 - a^3)^3}$$

SSC CGL 24/8/2021 (Afternoon)

(a) $\frac{5}{2}(a^3 + b^3)(b^3 + c^3)(c^3 + a^3)$

(b) $\frac{5}{2}(a^3 + b^3)(b^3 - c^3)(c^3 - a^3)$

(c) $\frac{5}{2}(a^3 - b^3)(b^3 + c^3)(c^3 + a^3)$

(d) $\frac{5}{2}(a^3 - b^3)(b^3 - c^3)(c^3 + a^3)$

If $a^2+b^2+c^2=6.25$, if $ab+bc+ca= 0.52$,
what is the value of $(a+b+c)$, if
 $(a+b+c)<0$?

± 2.7

± 28

-2.7

-2.8

If $x/(b+c-a)= y/(b-c-a)=z/(a+b-c)=k$,
Find $x^2+y^2+z^2-2xy-2yz+2zx$?

A. $K^2(a^2+b^2+c^2)$

B. $K^2(a^2-b^2+c^2)$

C. $K^2(a+b+c)^2$

D. $K^2(a-b+c)^2$

If $x = 32.5$, $y = 34.6$ and $z = 30.9$, then the value of $x^3 + y^3 + z^3 - 3xyz$ is $0.98k$, where k is equal to:

यदि $x = 32.5$, $y = 34.6$ और $z = 30.9$, तो $x^3 + y^3 + z^3 - 3xyz$ का मान $0.98k$ है, जहाँ x इसके बराबर है:

- 1033
- 933
- 1026
- 961

If $x=7$, find $x^7 - 8x^6 + 8x^5 - 8x^4 + 8x^3 - 8x^2 + 8x$

- 0
- 1
- 7
- 8

If $x = 78$, find $x^7 - 79x^6 + 79x^5 - 79x^4 + 79x^3 - 79x^2 + 78x + 1$?

0

1

78

79

If $x = 7$, find $x^6 - 8x^5 + 8x^4 - 8x^3 + 10x^2 - 8x + 13$.

a. 100

b. 102

c. 1074

d. 104

if $a+b+c=1$, $a^2+b^2+c^2=2$, $a^3+b^3+c^3=3$. Find abc ?

$1/6$

$1/3$

$-1/6$

$-1/3$

if $a+b+c=1$, $a^2+b^2+c^2=2$, $a^3+b^3+c^3=3$.
Find $a^4+b^4+c^4$?

Irrelevant (ratt lo)

$25/6$

$6/25$

$-25/6$

$-6/25$

If $\sqrt[3]{x+y} + \sqrt[3]{y+z} + \sqrt[3]{z+x} = 0$, find $x^3 + y^3 + z^3$.

- a. $\frac{1}{8}xyz$
- b. $(x+y)(y+z)(z+x)$
- c. $\frac{3}{8}(x+y)(y+z)(z+x)$
- d. $3xyz$

28.If $\sqrt[3]{x+y-z} + \sqrt[3]{y+z-x} + \sqrt[3]{z+x-y} = 0$,

find $x^3 + y^3 + z^3 - 27(x+y-z)(y+z-x)(z+x-y)$.

- a. $3(x+y)(y+z)(z+x)$
- b. $-3(x+y)(y+z)(z+x)$
- c. $-6(x+y)(y+z)(z+x)$
- d. $6(x+y)(y+z)(z+x)$

If $x + y + z = 0$, find $(x + y - z)^3 + (z + x - y)^3 + (y + z - x)^3$.

- a. $(x + y + z)^3$
- b. $3(x + y)(y + z)(z + x)$
- c. $24xyz$
- d. $-24xyz$

Find $(72)^3 - 23^3 - 24^3 - 25^3$?

$$3 \times 47 \times 49 \times 48$$

$$3 \times 23 \times 4 \times 25$$

$$3 \times 46 \times 48 \times 50$$

$$3 \times 57 \times 59 \times 58$$

If $x = \sqrt{2} - 1$, $y = 3/\sqrt{2}$

$$(x-2y+1)(x^2+4y^2+2x+2y+2xy+1)$$

- A. $15\sqrt{2}$
- B. $52\sqrt{2}$
- C. $-52\sqrt{2}$
- D. $-15\sqrt{2}$

If $x = \sqrt{2} - 2$, $y = 5/\sqrt{2}$

$$(x+2+8y/5)(x^2+4y^2+4x+4y+2xy+4)$$

- A. $310\sqrt{2}$
- B. $315\sqrt{2}$
- C. $311\sqrt{2}$
- D. $312\sqrt{2}$

If $x^{2a} = y^{2b} = z^{2c} \neq 0$ and $x^2 = yz$, find the value of $\frac{ab+bc+ca}{bc}$

यदि $x^{2a} = y^{2b} = z^{2c} \neq 0$ व $x^2 = yz$ हो, तो $\frac{ab+bc+ca}{bc}$ का मान ज्ञात करें।

- A. $3ac$
- B. $3ab$
- C. $3bc$
- D. 3

If $a^x = b^y = c^z$ and $b^2 = ac$, then the value of y is equal to

A $\frac{2xz}{x+z}$

B $\frac{xz}{2(x-z)}$

C $\frac{xz}{2(z-x)}$

D $\frac{2xz}{(x-z)}$

Thankyou

Algebra

Class -7

If $3^a = 27^b = 81^c$ and $abc = 144$, then the value of $12 \left(\frac{1}{a} + \frac{1}{2b} + \frac{1}{5c} \right)$ is:

- A. $33/10$
- B. $18/10$
- C. $17/120$
- D. $18/120$

If $x^2=17x+y$, $y^2=x+17y$, $x \neq y$, find $\sqrt{x^2+y^2+1}$

- A. 17
- B. 18
- C. 19
- D. 20

If $x^2=b-ax$, $x^3(x^3+c)=d$, which of the following can be the value of c and d ?

- 1. $C=a^3-3ab$, $d=b^3$
- 2. $C=-a^3-3ab$, $d=b^3$
- 3. $C=a^3+3ab$, $d=b^3$
- 4. $C=a^3+3ab$, $d=-b^3$

HOW TO USE BOUNDARY CONDITIONS ?

If/यदि $(x - y + 8)[xy - 8(x - y)] - 8xy = 0, x \neq -8, y \neq 8$, then/तो $\sqrt{\frac{16 + (x - y)^3}{2(x - y)^2 + 1}} = ?$

A. 20

B. 16

C. 4

D. none

If/यदि $(x + y + 7)[xy + 7(x + y)] - 7xy = 0, x \neq -7, y \neq -7$ then/तो $\frac{\sqrt{5}}{(x+y+\sqrt{5})^3} - \frac{1}{5} = ?$

- A. 2
- B. 0
- C. 1
- D. none

If $\frac{b-c}{a} + \frac{a+c}{b} + \frac{a-b}{c} = 1, a - b + c = 0$, then

- a. $\frac{1}{b} = \frac{1}{a} - \frac{1}{c}$
- b. $\frac{1}{a} = \frac{1}{b} + \frac{1}{c}$
- c. $\frac{1}{b} = \frac{1}{a} + \frac{1}{c}$
- d. None of the above

A and B are + integer, if $A+B+AB = 186$, find the positive difference between A and B ?
 $A, B \leq 20$

A और B + पूर्णांक हैं, यदि $A + A + AB = 186$, A और B के बीच सकारात्मक अंतर पाएं? A, $B \leq 20$

- 4
- 5
- 6
- 7

If $xyz + xy + yz + zx + x + y + z = 384$, where x, y, z are positive integers, then the value of $x + y + z$.

- a. 23
- b. 18
- c. 20
- d. 35

If $x + y + xy = 29$, $y + z + yz = 41$, $z + x + zx = 34$, find $3z - 4x + 5y$.

by *Abhas Saini*

24

27

28

26

If $A = 125$ and $B = 8$, then what is the value of $(A+B)^3 - (A-B)^3 - 6B(A^2 - B^2)$?

4096

4608

4224

3456



(VALUE PUTTING AND DEGREE CONCEPTS)

✓ EXCLUSIVE CONCEPT IN VALUE PUTTING.

If x, y, z and w are in proportion, Find $\frac{x^m + y^m + z^m + w^m}{x^{-m} + y^{-m} + z^{-m} + w^{-m}}$

a. $\frac{x}{y}$

b. $(xyzw)^{\left(\frac{m}{2}\right)}$

c. 1

d. $(xyzw)^m$

If q is the mean proportional between p and r ,

$$\text{find } \frac{p^2 - q^2 + r^2}{p^{-2} - q^{-2} + r^{-2}}$$

- a. $p^2 q^3$
- b. q^3
- c. q^4
- d. $p^2 r^2 q^4$

If $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, find $yx + yz + zx$.

- a. $\frac{(a+b+c)^2}{x^2+y^2+z^2}$
- b. $\frac{(x^2(a+b+c)^2 - a^2(x^2+y^2+z^2))}{2a^2}$
- c. $\frac{ax+by+cz}{(a+b+c)^2}$
- d. 3

If $\frac{P}{p} = \frac{Q}{q} = \frac{R}{r} = \frac{S}{s}$, $p + q + r + s \neq 0$,

find $\sqrt{Pp} + \sqrt{Qq} + \sqrt{Rr} + \sqrt{Ss}$.

a. $\sqrt{(P + Q + R + S)(p + q + r + s)}$

b. $\frac{\sqrt{P+Q+R+S}}{p+q+r+s}$

c. $\frac{P+Q+R+S}{\sqrt{(p+q+r+s)}}$

d. $(p + q + r + s)(\sqrt{P} + \sqrt{Q} + \sqrt{R} + \sqrt{S})$

$\frac{a^3(b+c)}{(a-b)(a-c)} + \frac{b^3(c+a)}{(b-c)(b-a)} + \frac{c^3(a+b)}{(c-a)(c-b)}$ is equal to

a. abc

b. $a + b + c$

c. $ab + bc + ca$

d. 3

$(bc + ca + ab)^3 - b^3c^3 - c^3a^3 - a^3b^3$ is equal to

- a. $3abc(a + b)(b + c)(c + a)$
- b. $(a + b)(b + c)(c + a)$
- c. $(a - b)(b - c)(c - a)$
- d. $24abc$

$a(b - c)^2 + b(c - a)^2 + c(a - b)^2 + 8abc =$

- a. $(a + b)(b + c)(c + a)$
- b. $(a - b)(b - c)(c - a)$
- c. 0
- d. abc

$a(b - c)^3 + b(c - a)^3 + c(a - b)^3$ is equal to

- a. $(a + b)(b + c)(c + a)(a + b + c)$
 - b. $(a - b)(b - c)(c - a)(a + b + c)$
 - c. $(a - b)(b - c)(c - a)$
 - d. $6abc$
-

$$(a + b + c)^4 - (a + b)^4 - (b + c)^4 - (c + a)^4 + a^4 + b^4 + c^4 =$$

- a. $12abc(a + b + c)$
- b. 36
- c. abc
- d. $abc(a + b + c)$

$$(a + b + c)^3 - (a + b - c)^3 - (b + c - a)^3 - (c + a - b)^3 =$$

- a. $8(a + b + c)$
- b. $24abc$
- c. $12abc$
- d. 24

$$\frac{2a}{a+b} + \frac{2b}{b+c} + \frac{2c}{c+a} + \frac{(a-b)(b-c)(c-a)}{(a+b)(a+c)(b+c)} \text{ is equal to}$$

- a. 1
- b. 2
- c. 3
- d. 4

Find $(x - a)(x - b)(x - c)$.

a. $x^3 - (a + b + c)x^2 + (bc + ca + ab)x - abc$

b. $x^3 + (a + b + c)x^2 + (bc + ca + ab)x + abc$

c. $x^3 - (a + b + c)x^2 + (a + b + c)x - abc$

d. $x^3 - (a + b + c)x^2 - (a + b + c)x - abc$

Find the factors of $(a^2 + 4b^2 + 4b - 4ab - 2a - 8)$.

a. $(a - 2b - 4)(a - 2b + 2)$

b. $(a - b + 2)(a - 4b - 4)$

c. $(a + 2b - 4)(a + 2b + 4)$

d. $(a + 2b - 1)(a - 2b + 1)$

$$(a + b + 2c)(a^2 + b^2 + 4c^2 - ab - 2bc - 2ca) =$$

a. $a^3 + b^3 + 8c^3 - 6abc$

b. $a^3 + b^3 + 6c^3 - 6abc$

c. $a^3 + b^3 + 8c^3 - 2abc$

d. $a^3 + b^3 + 8c^3 - abc$

$$(a + b - c)^3 + (a - b + c)^3 - 8a^3 =$$

a. $6a(a - b + c)(c - a - b)$

b. $6a(a + b - c)(a - b + c)$

c. $3a(a + b - c)(a - b + c)$

d. $3a(a - b + c)(c - a + b)$

If $x = 3-4y$, then $(x^3 + 64y^3 + 36xy)$ is equal to:

यदि $x = 3 - 4y$ है, तो $(x^3 + 64y^3 + 36xy)$ का मान क्या होगा ?

- (a) $27(1+3xy)$
- (b) $9(3+5y)$
- (c) $27(1+xy)$
- (d) 27

If $2z = x + y$, then the value of $x/(x-z) + y/(y-z)$?

यदि $2z = x + y$, तो $x/(x-z) + y/(y-z)$ का मान ज्ञात करें ?

- (a) 0
- (b) 1
- (c) 2
- (d) 5

If $x + y + z = 0$, then find the value of $[(x^2)/(x^2 - yz)] + [(y^2)/(y^2 - zx)] + [(z^2)/(z^2 - xy)]$

यदि $x + y + z = 0$ हो तो $[(x^2)/(x^2 - yz)] + [(y^2)/(y^2 - zx)] + [(z^2)/(z^2 - xy)]$

- (a) 2
- (b) 1
- (c) 3
- (d) 0

If $a + b + c = 0$ then find the value of $(a^2 + b^2 + c^2)/(a^2 - bc)$

$a + b + c = 0$, $(a^2 + b^2 + c^2)/(a^2 - bc)$ का मान ज्ञात करें?

- (a) -1
- (b) 1
- (c) 2
- (d) -2

If $x - y + z = 0$, then find the value of $(y^2/2xz) - (x^2/2yz) - (z^2/2xy)$

यदि $x - y + z = 0$ है, तो $(y^2/2xz) - (x^2/2yz) - (z^2/2xy)$ का मान ज्ञात कीजिए ।

- (a) $3/2$
- (b) $1/2$
- (c) -6
- (d) $-3/2$

If $ab(a+b) + bc(b+c) + ca(c+a) + 3abc$, If

$a+b+c=13$ and $ab+bc+ca = 5$

- a) 23
- b) 13
- c) 65
- d) 5

If $a + b + c = 6$ and $a^2 + b^2 + c^2 = 38$,

find the value of $a(b^2 + c^2) + b(c^2 + a^2) + c(a^2 + b^2) + 3abc$.

1

0

6

-6

**If $a+b+c=12$, $a^2+b^2+c^2=134$, find
 $bc(b+c)+ca(c+a)+ab(a+b)+3abc$?**

30

60

90

120

If $x + y + 3 = 0$, then find the value of $x^3 + y^3 - 9xy + 9$.

यदि $x + y + 3 = 0$ है, तो $x^3 + y^3 - 9xy + 9$ का मान ज्ञात कीजिए ।

- (a) -18
- (b) -36
- (c) 18
- (d) 36

If $x = 3 - 4y$, then $(x^3 + 64y^3 + 36xy)$ is equal to:

यदि $x = 3 - 4y$ है, तो $(x^3 + 64y^3 + 36xy)$ का मान क्या होगा ?

- (a) $27(1+3xy)$
- (b) $9(3+5y)$
- (c) $27(1+xy)$
- (d) 27

If $2z = x + y$, then the value of $x/(x-z) + y/(y-z)$?

यदि $2z = x + y$, तो $x/(x-z) + y/(y-z)$ का मान ज्ञात करें ?

- (a) 0
- (b) 1
- (c) 2
- (d) 5

If $a + 3/b = 3$, $b + 1/c = 1$

Find $a(c-1) - abc$?

- 0
- 1
- 1
- 2

Which among the below option can be a square root of?

निम्नलिखित का वर्गमूल क्या है?

$$(x^2 - 14x + 49)(x^2 + 6x + 9)$$

- (a) $(x - 4)(x + 9)$
- (b) $(x - 1)(x + 17)$
- (c) $(x - 7)(x + 3)$
- (d) $(x - 3)(x + 8)$

If $x - y + z = 0$, then find the value of $(y^2/2xz) - (x^2/2yz) - (z^2/2xy)$

यदि $x - y + z = 0$ है, तो $(y^2/2xz) - (x^2/2yz) - (z^2/2xy)$ का मान ज्ञात कीजिए ।

- (a) $3/2$
- (b) $1/2$
- (c) -6
- (d) $-3/2$

• If $x + y = 36$, then find $(x - 27)^3 + (y - 9)^3$
 यदि $x + y = 36$ है, तो $(x - 27)^3 + (y - 9)^3$ ज्ञात कीजिए।

•

a) 1

b) 81

c) $2y$

d) 0

If $\frac{a^2 + b^2 + c^2 - 1024}{ab - bc - ca} = -2$ and $a + b = 5c$, where $c > 0$, then the value of c is _____.

अगर $\frac{a^2 + b^2 + c^2 - 1024}{ab - bc - ca} = -2$ तथा $a + b = 5c$, जहाँ $c > 0$, तो c का मान है _____.

a) 8

b) 4

c) 12

d) 5

If $x^2 + y^2 + z^2 = xy + yz + xz$ and $x = 1$, then

find the value of $\frac{10x^4 + 5y^4 + 7z^4}{13x^2y^2 + 6y^2z^2 + 3x^2z^2}$.

यदि $x^2 + y^2 + z^2 = xy + yz + xz$ तथा $x = 1$, तो
 $\frac{10x^4 + 5y^4 + 7z^4}{13x^2y^2 + 6y^2z^2 + 3x^2z^2}$ का मान ज्ञात कीजिए।

a) 2

b) 0

c) -1

d) 1

If $mx^m - nx^n = 0$, then what is the value of

$\frac{1}{x^m + x^n} + \frac{1}{x^m - x^n} + 1$ terms of x^n is _____,

where x, m, n are > 0

यदि $mx^m - nx^n = 0$ है, तो $\frac{1}{x^m + x^n} + \frac{1}{x^m - x^n}$ का मान क्या है x^n के संदर्भ में? जहाँ $x, m, n > 0$ हैं

a) $2mn/(x^n(n^2 + m^2))$ b) $2mn/(x^n(m^2 - n^2))$

c) $2mn/(x^n(n^2 + m^2))$ d) $2mn/(x^n(n^2 - m^2))$

If $a + \frac{3}{b} = 3$, $b + \frac{1}{c} = 1$

Find $a(c-1) - abc$?

0

1

-1

2

Exclusive Concept !!!

$$\text{If } \frac{x}{(a-b)(a+b-2c)} = \frac{y}{(b-c)(b+c-2a)} = \frac{z}{(c-a)(a+c-2b)},$$

find $x + y + z$.

- a. abc
- b. $a + b + c$
- c. 1
- d. 0

If $x^3 + y^3 + z^3 = m^3$, $x^2 + y^2 + z^2 = m^2$ and $x + y + z = m$, find xyz .

यदि $x^3 + y^3 + z^3 = m^3$, $x^2 + y^2 + z^2 = m^2$ व $x + y + z = m$ हो, तो xyz ज्ञात करें।

- A. -1
- B. $m^2 - 1$
- C. 0
- D. $m^2 + 1$



$$\frac{1}{bc(a-b)(a-c)} + \frac{1}{ca(b-c)(b-a)} + \frac{1}{ab(c-a)(c-b)} = ?$$

- A. $a + b + c$
- B. 3
- C. $ab + \underline{bc} + ca$
- D. 0



If $a+b+c=0$, find $(1/a^2+b^2-c^2)+(1/b^2+c^2-a^2)+(1/a^2+c^2-b^2)$

If $ab+bc+ca=abc$

Find $[(b+c)/bc(a-1)] + [(a+c)/ac(b-1)] + [(a+b)/ab(c-1)]$

$XYZ=4(4+X+Y+Z)$

Find $1/(2+X) + 1/(2+Y) + 1/(2+Z)$

1

$1/2$

$3/2$

$-1/2$

If $\frac{x-a^2}{b^2+c^2} + \frac{x-b^2}{c^2+a^2} + \frac{x-c^2}{a^2+b^2} = 3$, then $x = ?$

यदि $\frac{x-a^2}{b^2+c^2} + \frac{x-b^2}{c^2+a^2} + \frac{x-c^2}{a^2+b^2} = 3$ हो, तो $x = ?$

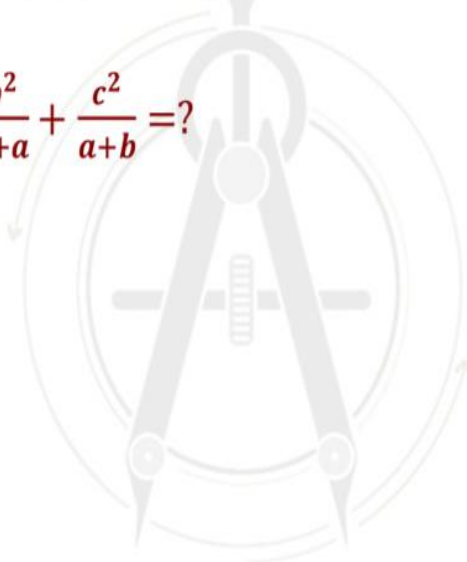
1. $a^2 + b^2 - c^2$
2. $a^2 + b^2$
3. $a^2 - b^2 - c^2$
4. $a^2 + b^2 + c^2$



If $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = 1$, then $\frac{a^2}{b+c} + \frac{b^2}{c+a} + \frac{c^2}{a+b} = ?$

यदि $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = 1$ हो, तो $\frac{a^2}{b+c} + \frac{b^2}{c+a} + \frac{c^2}{a+b} = ?$

1. 0
2. 1
3. 2
4. 3



If $\frac{x-a^2}{b+c} + \frac{x-b^2}{c+a} + \frac{x-c^2}{a+b} = 4(a+b+c)$, then $x = ?$

यदि $\frac{x-a^2}{b+c} + \frac{x-b^2}{c+a} + \frac{x-c^2}{a+b} = 4(a+b+c)$, हो, तो $x = ?$

1. $(a^2 + b^2 + c^2)$
2. $ab + bc + ca$
3. $(a^2 + b^2 + c^2 - ab - bc - ca)$
4. $(a + b + c)^2$

If $\frac{x}{a-x} + \frac{y}{b-y} + \frac{z}{c-z} = 2$, then $\frac{a}{a-x} + \frac{b}{b-y} + \frac{c}{c-z} = ?$

यदि $\frac{x}{a-x} + \frac{y}{b-y} + \frac{z}{c-z} = 2$ हो, तो $\frac{a}{a-x} + \frac{b}{b-y} + \frac{c}{c-z} = ?$

- 1.5
- 2.10
- 3.15
- 4.20

If $\frac{a}{2-3a} + \frac{b}{2-3b} + \frac{c}{2-ca} = 2$, then $\frac{1}{2-3a} + \frac{1}{2-3b} + \frac{1}{2-ca} = ?$

यदि $\frac{a}{2-3a} + \frac{b}{2-3b} + \frac{c}{2-ca} = 2$ हो, तो $\frac{1}{2-3a} + \frac{1}{2-3b} + \frac{1}{2-ca} = ?$

1.3.5

2.4.5

3.2.9

4.4.8

57 1 1 1 1 1

Maxima minima for Algebra

DIFFERENTIATION ROLE

At what value of x , $40 - x^2 + 5x$ will be the maximum value?

x के कौन से मान पर, $40 - x^2 + 5x$ अधिकतम मान होगा?

- 1) $185/4$
- 2) $4/185$
- 3) $(-185/4)$
- 4) $58/4$

Find the maximum value of $-x^2 - 8x + 7$.

$-x^2 - 8x + 7$ का अधिकतम मान ज्ञात कीजिए।

- 1) 7
- 2) 13
- 3) 23
- 4) 3

The minimum value of $(x - 2)(x - 9)$ is
 $(x - 2)(x - 9)$ का न्यूनतम मान क्या है?

- 1) $49/4$
- 2) 0
- 3) $11/4$
- 4) -49

Q. If $a + b + c + d = 2$, then the maximum value of $(1 + a)(1 + b)(1 + c)(1 + d)$?

(a) $91/9$ (b) $81/16$ (c) $63/22$ (d) $54/13$

If $(x + y + z = 18)$, then find the maximum value of $(x + 1)(y + 2)(z + 3)$.

यदि $(x + y + z = 18)$ है, तो $(x + 1)(y + 2)(z + 3)$ का अधिकतम मान होगा-

- 1) 216**
- 2) 512**
- 3) 6**
- 4) 18**

If $(x + y + z) = 27$, then what will be the maximum value of $(x - 5)(y + 4)(z + 7)$?

यदि $(x + y + z) = 27$ है, तो $(x - 5)(y + 4)(z + 7)$ का अधिकतम मान क्या होगा?

- 1) 1008
- 2) 832
- 3) 1331
- 4) 729

If $p + q + r + s = 22$, then what will be the maximum value of $[(1+p) / (q - 1)] + [(r - 1) / (1 + s)]$?

यदि $p + q + r + s = 22$ है, तो $[(1 + p) / (q - 1)] + [(r - 1) / (1 + s)]$ का अधिकतम मान क्या होगा?

- 1) 227/117
- 2) 235/117
- 3) 242/117
- 4) 250/117

Find the maximum value of
 $x^2 + \{1/(x^2 + 2)\}$

- 1) 0**
- 2) 13**
- 3) 5**
- 4) 3**

Find the minimum value of $(a + b) (1/a + 1/b)$
(Given a and b are positive.)

$(a + b) (1/a + 1/b)$ का न्यूनतम मान ज्ञात कीजिए (दिया है a और 5 धनात्मक हैं।)

- 1) 8**
- 2) 2**
- 3) 4**
- 4) 1**

If $-5 \leq x \leq 1$ and $-1 \leq y \leq 5$, then the minimum value of $2y - 3x$ is:

अगर $-5 \leq x \leq 1$ एवं $-1 \leq y \leq 5$ है, तो $2y - 3x$ का न्यूनतम मान क्या होगा?

- 1) -5
- 2) -9
- 3) -8
- 4) -10

Q. If x and y are real numbers, then the least possible value of $4(x-2)^2 + (y-3)^2 - 2(x-3)^2$ is ?

- (a) -8 (b) -4 (c) 1 (d) 3

if $2a+3b=15$, find the maximum value of $a^3 \cdot b^2$, given a, b are positive real number ?

212

243

216

256

if $a \cdot b=6$, find the minimum value of $2a+3b$, a, b are the positive real number ?

10

12

15

16

If x and y are positive real numbers and $xy = 8$, then the minimum value of $2x + y$ is

यदि x और y धन वास्तविक संख्या हैं और $xy = 8$ हैं, तब $2x + y$ का न्यूनतम मान क्या होगा?

- 1) 9
- 2) 17
- 3) 10
- 4) 8

Thankyou